



MA.8.NSO.1.1

Benchmark

MA.8.NSO.1.1 **Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.**

Example: Within the expression $1 + \sqrt{30}$, the irrational number $\sqrt{30}$ can be estimated to be between 5 and 6 because 30 is between 25 and 36. By considering $(5.4)^2$ and $(5.5)^2$, a closer approximation for $\sqrt{30}$ is 5.5. So, the expression $1 + \sqrt{30}$ is equivalent to about 6.5.

Benchmark Clarifications:

Clarification 1: Instruction includes the use of number line and rational number approximations, and recognizing pi (π) as an irrational number.

Clarification 2: Within this benchmark, the expectation is to approximate numerical expressions involving one arithmetic operation and estimating square roots or pi (π).

Connecting Benchmarks/Horizontal Alignment

- MA.8.AR.2.3
- MA.8.GR.1.1, MA.8.GR.1.2

Terms from the K-12 Glossary

- Expression
- Irrational Numbers
- Number Line
- Pi
- Rational Numbers
- Real Numbers

Vertical Alignment

Previous Benchmarks

- MA.7.NSO.1.2

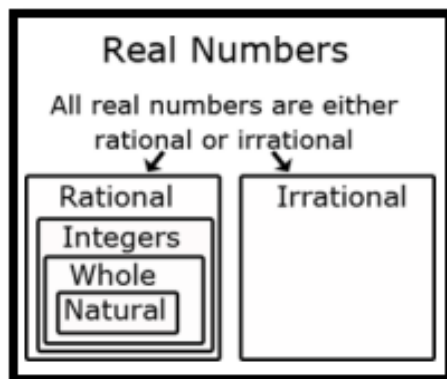
Next Benchmarks

- MA.912.AR.3.1

Purpose and Instructional Strategies

In grade 7, students expressed rational numbers using terminating and repeating decimals. In grade 8, students define irrational numbers and determine their approximate location on a number line. In Algebra 1, students will expand their knowledge of the real number system to understand that complex numbers exist from their work with quadratics.

- Instruction includes using and creating a graphic organizer to show the relationship between the subsets of the real number system.



- Once students understand that (1) every rational number has a decimal representation that either terminates or repeats, and (2) every terminating or repeating decimal is a rational number, they can reason that on the number line, irrational numbers must have decimal representations that neither terminate nor repeat.
- Students sometimes overgeneralize that all square roots are rational numbers concluding that irrational numbers are unusual and rare. Instruction includes a variety of examples of irrational numbers. Irrational numbers are much more plentiful than rational numbers, in the sense that they are denser in the real number line.
- Instruction includes the understanding that the square root of a whole number is either another whole number or is irrational. When the result is another whole number, the original whole number is a perfect square. This fact is particularly relevant when the Pythagorean Theorem is applied to find a missing side length of a triangle whose other two side lengths are whole numbers.
- Instruction includes the understanding that the cube root of a whole number is either another whole number or is irrational. When the result is another whole number, the original whole number is a perfect cube.
- Instruction includes the understanding that adding or subtracting a rational number and an irrational number produces an irrational number. The same is true of multiplication or division unless the rational number is 0.
- Students should develop estimating skills when working with square roots without the use of a calculator. One strategy is to use benchmark square roots to determine an approximate value.
 - For example, to find an approximation of $\sqrt{28}$, first determine the perfect squares 28 is between, which would be 25 and 36. The square roots of 25 and 36 are 5 and 6, respectively, so we know that $\sqrt{28}$ is between 5 and 6. Since 28 is closer to 25, an estimate of the square root would be closer to 5.

Common Misconceptions or Errors

- Students may incorrectly believe that pi (π) is a rational number since they have only been introduced to a decimal approximation and a fraction approximation. To address this misconception, instruction includes looking further at the decimal representation of pi (π) so that students will notice that a pattern will not emerge. In fact, a pattern never emerges, therefore, pi is irrational.
- Students may incorrectly think that the number line only has the numbers that are labeled.

- Students may incorrectly think a numerical expression that includes addition or subtraction cannot be placed on a number line.
 - For example, $2 + \sqrt{3}$ can be placed on a number line at approximately 3.73.

Strategies to Support Tiered Instruction

- Teacher provides opportunities to look at the decimal representation of pi (π) and comparing decimal representations of rational numbers and irrational numbers.
- Teachers provide opportunities for practice with performing decimal expansion of rational and irrational numbers to check for a pattern.
- Teacher models how to estimate a numerical expression with addition or subtraction and locates a place on the number line.
 - For example, show a representation of pi (π) and compare it to a decimal representation of $\frac{22}{7}$. Students should identify that $\frac{1}{3}$ is a rational number and repeats 0.3333, and that $\frac{22}{7}$ is pi ($\pi = 3.1415\dots$) and does not repeat.
 - For example, a first approximation of $3 + \sqrt{5}$, students could approximate $\sqrt{5}$ as 2.25 since 5 is between the perfect squares of 4 and 9, but closer to 4. Therefore $\sqrt{5}$ would be in between 2 and 3, but closer to 2. So, a reasonable guess for $\sqrt{5}$ can be 2.25 and therefore a reasonable estimate for $3 + \sqrt{5}$ would be $3 + 2.25$ which equals 5.25. If more accuracy is required, students should understand that a calculator is needed.
- Instruction includes looking further at the decimal representation of pi (π) so that students will notice that a pattern will not emerge. In fact, a pattern never emerges, therefore, pi is irrational.

Instructional Tasks

Instructional Task 1 (MTR.1.1)

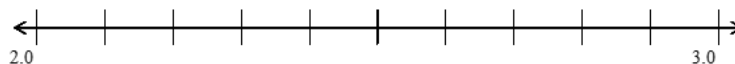
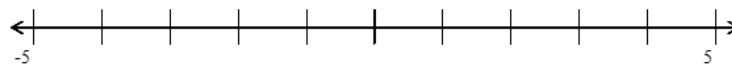
Part A. Provide an example of a rational number and explain how you can determine that it is rational.

Part B. Choose which number(s) below are irrational and explain how you can determine that they are irrational.

- a) $\sqrt{3} - 2$
- b) $6\sqrt{25}$
- c) $\sqrt[3]{36}$
- d) 2π
- e) $-4 + \sqrt[3]{-216}$

Instructional Task 2 (MTR.4.1)

Part A. Use the number lines below to estimate the value of $\sqrt{8}$. Explain why you put the points where you did.



Part B. Plot $1 + \sqrt{8}$ on a number line. Explain your process with a partner.

Instructional Items

Instructional Item 1

Plot the following numbers on a number line showing their approximate location to the nearest hundredth.

- $\pi - 2$
- $-\left(\frac{1}{2}\pi\right)$
- $2\sqrt{2}$
- $2 + \sqrt{17}$

Instructional Item 2

Is 0.12345 a rational or irrational number? Explain your answer.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*